Solve the following recurrence for $n, k \geq 0$:

$$Q_{n,0} = 1 \quad Q_{0,k} = [k = 0];$$
$$Q_{n,k} = QQ_{n-1,k} + Q_{n-1,k-1} + \binom{n}{k}, \quad \text{for } n,k > 0.$$